Youssef Boutaleb

Data Engineer

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EU Visa Holder. Open to relocation within the EU and to fully remote roles.

Summary

Data Engineer with over two years of production experience building enterprise-scale ETL/ELT pipelines, API-led integrations and observability on Microsoft Azure. At JACQUEMUS I sit in the Data Platform team and own the retail order path end to end across order management and omnichannel inventory, for e-commerce, boutique and wholesale channels, on a Microsoft Fabric medallion lakehouse orchestrated with Fabric Data Factory and transformed with PySpark. I also lead the platform's FinOps and cloud cost governance. Certified by Microsoft (DP-300, DP-600, DP-700), Astronomer, MuleSoft, Talend and Datadog. Alongside the role I lecture on the M.Sc. Data Science programme at IHEC Sfax, teaching the two Data Engineering courses and the Machine Learning course, and I publish: one peer-reviewed paper in *Computers & Industrial Engineering* (Elsevier, 2025) and one under review.

Proof of impact:

- Cut Azure infrastructure spend by **€1,400 per month** by automating development-environment shutdowns with PowerShell runbooks in Azure Automation Accounts, matching compute size to each environment and redesigning shared Spark pools, with no SLA impact on morning business reporting.
- Brought **150+ production jobs** under end-to-end alert coverage by instrumenting ingestion workloads with Datadog APM and custom dashboards across Azure Data Factory, Microsoft Fabric and Spark execution pools, surfacing silent job and API failures before business-hours reporting windows.
- Guaranteed **zero feedback data loss** during third-party API outages on the Kenzo Paris / LVMH integration, using RAML contracts, Anypoint MQ queues and a 5-stage exponential backoff policy that separates transient retries from permanent dead-letter routing.
- Delivered a **100× signal-preprocessing speedup** by replacing a C# Savitzky-Golay implementation with optimized C++, the path that then went inline at approximately **2,000 frames per second** for real-time metal-alloy sorting.
- Merged **2 plugins** into the official Kanboard plugin directory (PR #586 and PR #585), one of them maintained at PHPStan Level 8 with 770 automated tests.

Experience

Data Engineer | JACQUEMUS | Paris, France | Aug 2024 - Present

Permanent | Remote | Luxury E-commerce & Retail | over 20 pipelines

JACQUEMUS is a French luxury fashion house selling ready-to-wear, leather goods, and accessories through its own e-commerce, its boutiques, and wholesale partners. Part of the Data Platform team, owning the retail order path end to end across OMS (order management system) and OCI (omnichannel inventory) for e-commerce, boutique, and wholesale channels. I design and operate the surrounding financial, product, pricing, warehousing, and invoicing pipelines on a Microsoft Fabric medallion lakehouse, while leading platform FinOps and cloud cost governance.

Data Engineering

- Built and maintained **over 20 enterprise pipelines** across Salesforce, ORLI/WMS, REST APIs, SFTP file shares, Azure Data Lake, and Azure SQL (SQL Server).
  Business impact: The analytics, logistics, e-commerce reporting, and inventory-management teams work from these flows, so a stalled pipeline delays morning commercial reporting and inventory synchronization.
- Engineered the platform on the **modern Azure data stack**: a **medallion lakehouse on Microsoft Fabric**, orchestrated with **Fabric Data Factory** and transformed with **PySpark notebooks**, landing order, financial, product, and pricing data across bronze, silver, and gold Delta layers.
  Business impact: Establishes an enterprise single source of truth with structured data layers, accelerating downstream Power BI analytics and cross-departmental reporting.
- Integrated live **Quarkus REST API services behind Azure Application Gateway** alongside scheduled batch jobs for real-time order and inventory events.
  Business impact: Bypasses batch refresh windows for high-priority telemetry, moving time-sensitive retail updates directly over low-latency API endpoints.

Cloud & FinOps

- Reduced **Azure infrastructure spend by €1,400 per month** by automating development-environment shutdowns with **PowerShell** runbooks in Azure Automation Accounts, managing Azure IAM access controls, matching compute size to each environment, and redesigning shared Spark pools.
  Business impact: Delivers a recurring monthly saving on the platform budget with zero SLA impact on morning business reporting pipelines.

Monitoring & Observability

- Instrumented ingestion workloads with **Datadog APM and custom monitoring dashboards** across Azure Data Factory, Microsoft Fabric, and Spark execution pools.
  Business impact: Provides end-to-end alert coverage across **150+ production jobs**, surfacing silent job or API failures before business-hours reporting windows.

- Authored platform architecture runbooks, disaster recovery procedures, and operational playbooks for the **150+ production pipelines** and the retail order path.
  Business impact: Empowers on-call engineers to recover failed runs directly from standard playbooks, shortening mean time to recovery (MTTR) during incident responses.

Stack: Apache Spark, Microsoft Fabric, Azure Data Factory, Datadog

Data Engineer | OLIVESOFT | Sfax, Tunisia | Feb 2024 - Jul 2024

End-of-studies Internship | On-site | Luxury E-commerce & Retail

OLIVESOFT is a Paris-headquartered data consultancy delivering data engineering, integration, and BI services to over 20 clients, primarily in luxury retail, spanning platform delivery, consulting, and end-to-end operational support. Engineered the production customer-feedback integration pipeline for the Kenzo Paris / LVMH account, building a 3-layer event-driven MuleSoft architecture on CloudHub between Salesforce Service Cloud and Diduenjoy.

- Architected an asynchronous 3-layer **MuleSoft API-led connectivity framework** (System and Process APIs) on **MuleSoft CloudHub**, using **DataWeave 2.0** to transform Salesforce order event triggers into Diduenjoy survey payloads.
  Business impact: Decoupled agent workflows and achieved a 42% customer survey response rate in the initial monthly report, establishing an NPS 48 baseline toward Kenzo's 55 NPS target.
- Implemented **RAML 1.0 REST contracts, Anypoint MQ queues, and a 5-stage exponential backoff retry policy** distinguishing transient retries from permanent routing to Dead-Letter Queues (DLQ).
  Business impact: Guaranteed zero feedback data loss during third-party API outages, with automated CloudHub PaaS notifications handling failure alerts.

Stack: MuleSoft CloudHub, DataWeave 2.0, Salesforce Service Cloud, Anypoint MQ, RAML, Diduenjoy

Machine Learning Engineer | REGIM Lab | Sfax, Tunisia | Jun 2023 - Aug 2023

Summer Internship | On-site | Time-Series Sensor Data (Energy Research) | 300 households

REGIM-Lab (REsearch Groups in Intelligent Machines) is the applied-research laboratory at the National Engineering School of Sfax (ENIS), developing intelligent systems, machine learning models, and sensor data architectures. Engineered a research prototype for secure, decentralized energy monitoring, ingesting smart-meter time-series telemetry from Australia's Ausgrid dataset to benchmark regression anomaly detection models and log tamper-proof cryptographic audit trails.

- Ingested and preprocessed **1 year of 30-minute interval telemetry (~5.2M records)** across 300 solar households with **Pandas and NumPy**, delivering live metrics via a **Flask, Jinja2, and JavaScript dashboard**.
- Engineered time-series features and benchmarked **Scikit-learn** regression models to flag meter anomalies, writing cryptographic hashes to **Solidity smart contracts via Web3.py** on a local Ganache testnet.

Business impact: Formed the core data architecture and empirical benchmarks for a peer-reviewed research paper published in [Computers & Industrial Engineering (Elsevier, 2025)].

Stack: Scikit-learn, Pandas & NumPy, Flask & Jinja2, Solidity & Web3.py, Ganache, Ausgrid Dataset

Machine Learning & Signal Processing Engineer | OEM ENGINEERING S.A.R.L | Sfax, Tunisia | Jun 2022 - May 2023

On-site | Time-Series Sensor Data (Industrial Sorting)

OEM ENGINEERING is a Canadian industrial engineering company headquartered in Sherbrooke, Quebec, with its engineering site in Sfax, Tunisia, building high-speed material-sorting systems that identify metal alloys inline from visible near-infrared (VNIR) camera signals. Engineered the real-time signal processing and inline material classification stack across two roles, building high-speed C++ filtering at ~2,000 FPS and offline XGBoost spectral model benchmarks.

Machine Learning & Statistical Engineer | Part-time | Sep 2022 - May 2023

Stack: C++, XGBoost, scikit-learn, NumPy

- **Real-Time Classification.**
  - Integrated the deployed **C++ Savitzky-Golay filtering path** into inline metal-alloy classification, sustaining approximately **2,000 frames per second** (<0.5 ms per frame latency budget) for real-time sorting decisions.
    Business impact: Alloy identification runs at full conveyor speed, meeting the strict hardware throughput budget required for inline sorting.
- **Model R&D.**
  - Engineered spectral feature baselines and evaluated offline **XGBoost models using Scikit-learn and NumPy** across a **~10,000+ VNIR sample dataset**, achieving **94.7% accuracy** across multi-class metal alloys while maintaining production separation due to offline latency constraints.

Signal Processing Engineer | Summer Internship | Jun 2022 - Aug 2022

Stack: C++, Savitzky-Golay Filtering, VNIR Spectroscopy

- Delivered a **100× signal-preprocessing speedup** by replacing a C# Savitzky-Golay implementation with optimized C++ filtering using fixed in-memory parameters and GCC optimization, while retaining the same filter behavior for inline sorting.
  Business impact: Preprocessing stopped being the bottleneck, and this is the path that went inline the following autumn, where the filter has to keep pace with the camera rather than catch up behind it.

Technical Skills

Category	Skills
Data Pipelines & Orchestration	Talend Data Integration, MuleSoft, Apache Airflow 3 (DAG authoring, retries, backfills), Azure Data Factory, Fabric Data Factory
Distributed Processing	Apache Spark, PySpark, Microsoft Fabric notebooks, Spark pool sizing and execution-plan tuning
Cloud Platform & FinOps	Microsoft Azure (Data Lake, Azure SQL, Application Gateway, Automation Accounts, IAM), Microsoft Fabric, PowerShell runbooks, cost governance and right-sizing
Data Warehousing & Modeling	Medallion architecture (bronze, silver, gold Delta layers), star schema and dimensional modeling, Azure SQL, SQL Server, PostgreSQL
Integration & APIs	MuleSoft API-led connectivity, CloudHub, DataWeave 2.0, RAML 1.0, Anypoint MQ, dead-letter queues, REST, Quarkus REST services, SFTP, Salesforce Service Cloud
Monitoring & Observability	Datadog (APM, distributed tracing, log management, dashboards, alerting), Log4j2, Loguru, runbooks and disaster recovery procedures
Programming & Scripting	Python, SQL, Java, PowerShell, C++, PHP, JavaScript
Machine Learning & Data Science	scikit-learn, XGBoost, TensorFlow, Keras, YOLOv8, pandas, NumPy, Matplotlib
Delivery & Collaboration	Git, GitHub Actions, Docker, DVC, Postman, FastAPI, Agile/Scrum, Jira, Confluence

Certifications

- **Microsoft**
  - Microsoft Certified: Azure Database Administrator Associate (DP-300)

    - [Microsoft Certified: Fabric Data Engineer Associate (DP-700)]
    - [Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)]
- **Astronomer**
    - Astronomer Certification DAG Authoring for Apache Airflow 3
    - Astronomer Certification for Apache Airflow 3 Fundamentals
- **MuleSoft**
    - MuleSoft Certified Developer Level 1
- **Talend**
    - Talend Data Integration Certified Developer
- **Datadog**
    - Datadog Certified: Fundamentals
    - Datadog Certified: APM & Distributed Tracing Fundamentals
    - Datadog Certified: Log Management Fundamentals

Education

Computer Science Engineer's Degree | National Engineering School of Sfax | Sfax, Tunisia | 2021 - 2024

Data Engineering & Distributed Systems | EUR-ACE® Accredited

Computer science engineering program specializing in Data Engineering, Distributed Systems, Cloud Architectures, and Applied AI.

Graduation Thesis

- **Subject:** fault-tolerant integration between a Salesforce service desk and an external customer-feedback platform, studied over a six-month final-year placement at OLIVESOFT.
- **The argument:** queued, asynchronous exchange keeps the service desk working when the downstream platform is slow or unavailable. The report set out the architecture, the failure cases it covers, and the trade-offs it accepts.
- **Outcome:** a written thesis and a running production integration, defended for the degree in 2024.

Data & Systems Specialization

- **Data engineering & cloud:** big data processing, database management systems (SGBD), business intelligence, parallel & distributed computing, and cloud architectures.
- **Applied AI:** machine learning and computer vision.
- **Software craft & systems:** DevOps, microservices, industrial blockchain, and clean coding.

Student Clubs & Training

- **Training Lead, Securinets ENIS:** ran the club's technical training track, 6 sessions a year (3 per semester), from x86 assembly through hands-on reverse-engineering labs to web exploitation and SQL injection, open to club members and to any ENIS student who turned up.
- **Finding the people to teach it:** brought in senior students and working engineers for the topics the club could not cover itself, and built the term's schedule around who was available.
- **Member, IEEE Student Branch ENIS:** part of the group running the introductory programming series (CS fundamentals, Python, C++ for competitive programming) .

Bachelor of Engineering (Mathematics and Physics) | Higher Institute of Applied Sciences and Technology of Mahdia | Mahdia, Tunisia | 2019 - 2021

National Preparatory Programme

The Tunisian national engineering preparatory program, MP track: intensive mathematics and physics, with admission to the National Engineering School of Sfax (ENIS) decided by competitive national ranking.

- **Advanced mathematics:** real and complex analysis, linear algebra, matrix theory, differential equations, numerical methods, and probability theory.
- **Physics & computational foundations:** classical mechanics, electromagnetism, thermodynamics, wave physics, and algorithmic problem solving.

Teaching

University Lecturer, IHEC Sfax (Institute of Higher Commercial Studies of Sfax), teaching on-site on the M.Sc. Data Science programme since 2024. Each course carries 32 h: 20 h of lectures across 5 modules, 8 h of hands-on labs (1 per module), and a 4 h capstone workshop. Delivery is bilingual, with lectures in French, technical terms in English, and all materials (slides, code repositories, assignments, exams) in English. Assessment: final exam 50%, module homework 20%, final project 15%, attendance and participation 15%.

Data Engineering 2 | M.Sc. Data Science | Fall 2025 | 32 h | 12 students

The second-year sequel, moving from building a pipeline to running one that survives contact with production failure modes.

- **Synchronous & Asynchronous Data Flows**
  - **Lab:** the same ingestion pipeline implemented twice (blocking HTTP vs. decoupled queue), then stalled under synthetic load to compare record loss against processing latency.
- **Resilient Pipeline Design & Idempotency**

    - **Lab:** a multi-task **Airflow** DAG authored with task-level retries, exponential backoff, and idempotent UPSERTs, then made to fail mid-run and repaired with a dated backfill over the missing window.
- **Processing at Volume with Apache Spark**
    - **Lab:** a join-and-aggregate transformation written in **Spark** against **PostgreSQL** source tables, refactored once the execution plan revealed shuffle bottlenecks, and the execution times compared.
- **Logging, Monitoring & Observability**
    - **Lab:** a pipeline instrumented with structured **Loguru** records shipped to **Datadog**, with an automated failure alert triggered and verified under synthetic error injection.
- **Packaging & Delivery**
    - **Lab:** a pipeline packaged into a **Docker** image with **DVC** data tracking and a **GitHub Actions** workflow that builds, tests, and publishes containers on main-branch commits.
- **Capstone: Fault-Tolerant Ingestion Pipeline**
    - **Technical Architecture & Data Flow:** end-to-end ingestion pipeline carrying data from a **PostgreSQL** operational source through a **Spark** transformation into an analytical target, scheduled as an **Airflow** DAG with retries and a working backfill path.
    - **Operability, Containerization & Monitoring:** structured **Loguru** logging surfaced on a **Datadog** dashboard, the complete pipeline containerized with **Docker**, and the input dataset versioned under **DVC**.
    - **Written Report & Oral Defense:** single-command execution repository accompanied by an architecture diagram and oral defense of retry, timeout, and alert threshold choices.

Data Engineering 1 | M.Sc. Data Science | Spring 2024 | 32 h | 16 students

The first-year data engineering course: how analytical data is stored, moved, and processed at a volume that no longer fits one machine.

- **Big Data Fundamentals & Memory Limits**
    - **Lab:** the same aggregation run over a dataset held whole in memory versus one read and partitioned in chunks, timed on both to locate the single-machine memory threshold.
- **Data Warehousing & Dimensional Modeling**
    - **Lab:** a star schema designed from a normalised operational source and created in **PostgreSQL**, then queried to measure analytical query performance gains.
- **ETL Architectures & Safe Transformations**
    - **Lab:** a **Python** ETL job extracting from an operational database, conforming records, and loading them into a **PostgreSQL** star schema without introducing duplicate rows.

- **Processing at Volume with Apache Spark**
  - **Lab:** rewriting Python ETL transformations into **Spark** DataFrame operations against larger extracts to identify the crossover point where distributed compute pays for itself.
- **Analytics & Reporting**
  - **Lab: Power BI** connected to a **PostgreSQL** star schema with a dashboard constructed over conformed dimensions and explicit semantic measures.
- **Capstone: Source-to-Dashboard Warehouse**
  - **Technical Architecture & Data Flow:** end-to-end data pipeline from an operational source to a published **Power BI** dashboard, featuring **Python** extraction, **Spark** transformations, and **PostgreSQL** serving.
  - **Operability & Data Modeling:** star schema implementation with explicit grain definitions, surrogate keys, and atomic transformation pipelines.
  - **Written Report & Oral Defense:** delivery of schema definitions, ETL code, and oral defense of dimensional modeling decisions (grain, keys, and normalization trade-offs).

Machine Learning | M.Sc. Data Science | Fall 2024 | 32 h | 12 students

The programme's applied machine learning course, taken from stating a deterministic domain problem as a learnable one to deciding whether the result means anything or the experiment has to be redesigned.

- **Supervised Learning & Imbalanced Classification**
  - **Lab:** twelve students in three teams, fitting one estimator per team (Decision Tree, Random Forest, Logistic Regression) over shared **Pandas** preparation in **Google Colab**, scoring accuracy vs. F1 across the room.
  - **Homework: Fraud Detection Benchmark.** An individual assignment in **Google Colab** where dataset and evaluation metrics are fixed while feature selection, estimator choice, and hyperparameter tuning are free.
- **Gradient Descent & Optimization (From Scratch)**
  - **Lab:** gradient descent implemented from the derivative upward in pure **NumPy**, tracing loss curves across learning rates until observing stall or divergence.
- **Unsupervised Learning & Clustering (From Scratch)**
  - **Lab:** K-Means implemented by hand in **NumPy** (assignment, centroid update, convergence test) and validated against **Scikit-learn**.
- **Model Evaluation & Diagnostic Curves**
  - **Lab:** a classifier cross-validated on an imbalanced target, with confusion matrices and **Matplotlib** learning curves used to diagnose underfitting vs. overfitting.
- **The Practical Workflow & Model Export**

- ○ **Lab:** an end-to-end Python workflow script (query ➔ prep ➔ fit ➔ evaluate) with model artifacts and metrics persisted together.
- **Capstone: Applied Prediction Study**
  - ○ **Technical Architecture & Data Flow:** supervised prediction study carried from a **PostgreSQL** extract through **Pandas** feature preparation into a tuned **Scikit-learn** model beating a naive baseline.
  - ○ **Operability & Pipeline Tuning:** cross-validated hyperparameter search over scikit-learn pipelines with persisted model metrics.
  - ○ **Written Report & Oral Defense:** notebook delivery featuring **Matplotlib** evaluation figures, defended on whether empirical results support the final conclusion.

Publications & Research

Secure and transparent energy management using blockchain and machine learning anomaly detection: A case study of the Ausgrid dataset

Computers & Industrial Engineering | Elsevier | 2025 | Published

Authors: N. Moumni, **Y. Boutaleb**, F. Chaabane, F. Drira

DOI: [10.1016/j.cie.2025.111040](https://doi.org/10.1016/j.cie.2025.111040)

Decentralized energy data architecture combining blockchain immutability for smart-meter audit trails with unsupervised machine learning pipelines to detect consumption anomalies.

- **Data pipeline & auditability:** ingests high-frequency smart-meter time-series into a decentralized ledger to ensure tamper-proof consumption records.
- **Unsupervised anomaly engine:** benchmarks data restoration and unsupervised anomaly annotation models over smart-meter streams to catch power theft and sensor anomalies.
- **Empirical validation:** benchmarked against 300 residential solar and grid consumption feeds from Australia's Ausgrid smart-meter dataset.

Dual verification-based multimodal approach for reliable pothole and speed bump detection on embedded system

Computers & Industrial Engineering | Under Review

Authors: S. Khemakhem, **Y. Boutaleb**, W. Ouarda

Edge-computing data pipeline combining accelerometer telemetry and vision streams with dual-verification consensus, preventing single noisy sensor channels from triggering false hazard alerts.

- **Multimodal sensor stream fusion:** synchronizes IMU motion telemetry with visual detection models on resource-constrained embedded targets.

- **Dual-verification consensus:** requires agreement between inertial and vision channels before registering a road anomaly, eliminating false positives from single-sensor noise.

Projects & Open Source

Kanboard Plugin: Draw.io Integration | Open source | Kanboard Plugin | 2026

Stack: PHP, JavaScript

Embeds **diagrams.net** into Kanboard Markdown with live in-browser editing, full-screen diagram viewing, and Wiki.js interop, accepted into the official Kanboard plugin directory.

- **Embedded diagram architecture:** renders fenced diagram blocks natively as sandboxed image elements across task descriptions, comments, and swimlanes without database schema modifications.
- **Interoperability & live editing:** integrates draw.io editor iframe messaging via postMessage with full-size diagram viewing and byte-for-byte Wiki.js format compatibility.
- **Security & CSP hardening:** enforces image-mode SVG rendering to neutralize active scripts, pins iframe origins, and restricts Content Security Policy to same-origin contexts.

Repository | Upstream PR #586, merged

Kanboard Plugin: File Attachment Interaction | Open source | Kanboard Plugin | 2026

Stack: PHP, JavaScript

Enterprise attachment framework adding in-modal previews (Office, PDF, CSV, code), live grid/text editors, and version branching, accepted into the official Kanboard plugin directory.

- **Multi-format parsing engine:** parses Office OpenXML (Word, PowerPoint, Excel), PDFs, Markdown, and source code directly inside task modals using pure-PHP DOM traversal.
- **Interactive editing & versioning:** provides in-browser spreadsheet grid editing, live text/code validation, and safe in-place updates or version branching for attachment revisions.
- **Defense-in-depth & verification:** neutralizes XXE attacks via LIBXML_NONET, restricts PDF iframe streaming with frame-ancestors CSP, and maintains 770 automated tests at PHPStan Level 8.

Repository | Upstream PR #585, merged

Custom YOLOv8 Model for Number Detection on Meters | Machine learning | Notebook | 2023

Stack: Python, YOLOv8, FastAPI, Gradio

Automates digit reading from utility meter photographs, serving trained weights through a containerized FastAPI REST service and a Hugging Face Gradio space.

- **End-to-end vision pipeline:** annotates custom utility meter datasets on Roboflow and trains YOLOv8 object detection weights across 100-epoch iterations.
- **Parallel accelerator acceleration:** speeds up feature extraction and bounding-box regression using Kaggle dual-accelerator parallel execution with DataParallel.
- **Containerized REST & Gradio serving:** packages trained weights into Dockerized FastAPI endpoints for automated image inference beside interactive Hugging Face Gradio spaces.

Repository | Live demo

Image Colorization with Deep Learning | Machine learning | Notebook | 2023

Stack: Python, TensorFlow, Keras

Reconstructs plausible color for grayscale images across four progressive notebook models (from baseline convolutional networks to adversarial GANs) to evaluate structural and color fidelity.

- **Progressive model architecture:** implements four sequential neural architectures (from baseline convolutional networks to generative adversarial networks) to reconstruct grayscale photos.
- **Perceptual loss & training:** optimizes generator networks using combined pixel loss and adversarial discriminator loss to balance structural consistency and vibrant colorization.
- **Fidelity evaluation:** evaluates color reconstruction across test datasets to measure structural similarity, chromatic accuracy, and visual artifact reduction.

Repository | Slides

Writing

Developing a Custom YOLOv8 Model for Number Detection on Meters Using FastAPI and Gradio | Medium | 2023

Walkthrough | 3,000 views, 1,500 reads as of Aug 2026

- Annotates the meter dataset in Roboflow and trains YOLOv8 from it.
- Trains across Kaggle's two TPUs with DataParallel.
- Serves the weights twice: a containerized FastAPI endpoint, and a Gradio app on Hugging Face.

Read the article

Simplify Your Log Management: Configuring DataDog Integration with Log4j2 In Mulesoft | Medium | 2024

Configuration Guide | 723 views, 424 reads as of Aug 2026

- A Log4j2 HTTP appender POSTs JSONLayout events straight to Datadog's Send Logs API.
- No agent on the host, and no collector in between.
- The appender is asynchronous, so logging never blocks a Mule execution thread.

Read the article

Workshops

Introduction to CS Fundamentals (Hardware) | OLIVESOFT | 2026

Workshop Series | Online | 2 h | Engineers & Executives | 30 participants

Two-part workshop series delivered for **OLIVE-LEARN** (OLIVESOFT's internal workshop training programme), tracing data movement from parallel hardware architectures down to logic gates.

- **Part 1: Efficient execution and parallel compute**
  - **Data & code execution:** CPU instruction pipelining, memory hierarchy (L1/L2/L3 caching), and how DMA, NICs, and APUs offload data movement from the main core.
  - **Parallel compute & Python bindings:** SIMD, SIMT, and MIMD execution models, demonstrating how high-level Python code (NumPy vectorization, PyTorch GPU threads) targets underlying vector hardware.
- **Part 2: Circuit-level computation**
  - Data storage and arithmetic built up from SR/D latches to 4-bit registers, adders, and an arithmetic logic unit.

Slides

Assembly Programming Workshop | Securinets ENIS | 2023

Hands-on Lab | On-site | 4 h | Engineering Students | 60 participants

A hands-on lab on hardware execution and low-level data handling, demonstrating the performance gap between high-level Python and bare-metal assembly.

- **Bus & memory architecture:** system bus data transfers, byte ordering (endianness), memory-mapped I/O, and register alignment.
- **ARM assembly & stack management:** register pressure, bitwise operations, stack frames, conditional execution, and guided coding exercises.

- **Python vs. ARM assembly benchmarking:** dual implementations showing how high-level abstractions introduce memory and stack overhead compared to direct register manipulation.
- **Bare-metal game build:** extension Breakout game implemented in x86 assembly, C, and C++ to compare manual memory control and execution speed.

Repository

Introduction to Competitive Programming | IEEE Student Branch ENIS | 2023

Workshop | On-site | 4 h | Engineering Students | 40 participants

A live-coding session guiding participants from brute-force thinking to optimized, pattern-based problem solving.

- **Core technique drills:** solved LeetCode problems live using prefix sums, binary search, and two-pointer strategies, converting $O(N^2)$ approaches into $O(N)$ and $O(\log N)$ solutions.
- **Complexity reasoning:** broke down Big-O time and space analysis for iterative and recursive/backtracking code, including theoretical lower bounds for sorting under different assumptions ($O(N \log N)$ vs. $O(N)$ when min/max are known).
- **Practice roadmap:** provided a self-study path through NeetCode, Codeforces, HackerRank, *Competitive Programming 4*, CLRS, and Sedgewick's Coursera algorithms course.

Introduction to Python | IEEE Student Branch ENIS | 2022

Workshop | Online | 2 h | Engineering Students | 12 participants

A foundational session that took students from Python syntax to parsing real datasets and visualizing them as charts.

- **Environment setup:** configured PyCharm with isolated virtual environments (`venv`) to establish clean, reproducible project workflows.
- **Core language & data structures:** covered Python syntax, dynamic state logic, and native structures (lists, tuples, dictionaries) as building blocks for organizing data.
- **File handling & visualization:** parsed CSV files using context managers (`with open`) and the `csv` module, then plotted the resulting data with Matplotlib.

Competitions & Awards

- **1st of 86 teams** | Hello World v4.0 | Regional | May 2024 (8 of 8 problems solved, team of 2, 4 h)
- **2nd of 52 teams** | Hello World v3.0 | Regional | May 2023 (6 of 8 problems solved, team of 2, 4 h)

- **13th of 90 teams** | Tunisian Collegiate Programming Contest 2023 | National Finalist | Oct 2023 (10 of 14 problems solved, team of 3, 5 h)
- **643rd of 7,094 teams** | IEEEXtreme Programming Competition 17.0 | International | Oct 2023 (11 of 26 problems solved, team of 2, 24 h)
- **Quarter-finalist of 200 teams** | A2SV (Africa to Silicon Valley) GenAI Hackathon | African | Aug 2023 (48 h)
  - Clinical records are written for clinicians. We built a service that turns one patient's diagnosis and medication history into a page they can read without one.
  - **The contract:** one row of patient data in, one plain-language HTML page out, with the risk factors and the advice arriving as structured JSON fields the template placed rather than prose it had to parse.
  - **Validation and fallback:** the shape was enforced rather than trusted, so every response was validated against the schema, a response that did not fit was retried, and exhausted retries fell back to a safe rendering.
    Result: unvalidated model output never reached a patient-facing page.
  - **Ownership:** the work split at that contract. Mohamed Brahim owned the prompt chain and the model call, and I owned the service, the schema and the rendering on the other side of it.
  - Stack: FastAPI, Pydantic, Jinja2, Hugging Face Inference API, Falcon 7B
- **21st of 80 teams** | Tunisian Collegiate Programming Contest 2022 | National Finalist | Oct 2022 (8 of 14 problems solved, team of 3, 5 h)
- **1,432nd of 6,376 teams** | IEEEXtreme Programming Competition 16.0 | International | Oct 2022 (8 of 26 problems solved, team of 2, 24 h)
- **7th of 21 teams** | Hello World v2.0 | Regional | May 2022 (5 of 8 problems solved, team of 2, 4 h)

Languages, Community & Continuing Education

Languages

- **Arabic:** Native.
- **French:** Full professional proficiency. Taught in French.
- **English:** Full professional proficiency. Taught in English, Published in English.

Community & Volunteering

- **Jeunes Ingénieurs de Djerba (JID) | Orientini | Djerba, Tunisia | Jul 2023.** Guided secondary school students through engineering fields and the careers each one leads to.
- **Jeunes Ingénieurs de Djerba (JID) | Orientini | Djerba, Tunisia | Jul 2022.** Guided secondary school students through engineering fields and the careers each one leads to.

- **Tunisian Red Crescent, Houmt Souk branch, Djerba | COVID-19 response | Djerba, Tunisia | Mar 2020 - Jun 2021.** Helped organise aid distribution and assisted with crowd coordination. Collected and distributed essential supplies, and supported elderly residents.

Continuing Education

- **Anthropic:** Claude Code 101, Claude 101, Claude Code in Action
- **OpenCV:** Vision Language Models (VLM) Bootcamp
- **Coursera:** Machine Learning Specialization: 3-course series by DeepLearning.AI, Build a Modern Computer from First Principles: From Nand to Tetris
- **Udemy:** Python and Flask Framework Complete Course, PyCharm Productivity and Debugging Techniques
- **DataCamp:** Data Analyst Associate
- **Scrumstudy:** Scrum Fundamentals Certified (SFC)